

2. The U.S. National Center for Health Statistics compiles natality (birth) statistics and publishes the results in *Vital Statistics of the United States*. Suppose that a random sample of 35 babies yields the following birth weights, in pounds:

7.2	6.0	8.6	4.5	2.0	7.9	4.0
2.4	5.9	7.3	7.3	7.0	6.3	8.1
7.0	7.3	6.6	5.2	9.8	8.0	10.9
6.1	3.8	5.0	8.0	10.7	9.7	6.0
6.5	10.3	7.6	6.5	7.1	5.8	6.9

The sample average ($\bar{x}$) here is 6.84 and the population standard deviation (σ) is 2.08.

(a) Use the data to obtain an estimate for the population average weight, μ , of all newborns. Is your estimate unbiased? Why?

(b) Find a 95% confidence interval for μ .

(c) Carefully interpret your result in part (b) in words.

(d) If you wished to estimate μ to within ± 0.25 lb, how many babies would you need in your sample?

(e) Suppose you read in a medical journal that the average weight of newborns in the U.S. is 8.0 lbs. Does this seem plausible in the light of the data above? Calculate the probability of getting a sample average as small as 6.84 lbs when the true population average is actually 8.0 lbs.

3. Discount stores often introduce new merchandise at a special low price in order to induce people to try it. However, a psychologist predicted that this practice would actually reduce sales. With the cooperation of a discount chain, an experiment was performed to test the prediction. Thirty-six pairs of stores were selected. A new kind of cookie was introduced in all 72 stores. Within each pair of stores, one was chosen at random to introduce the cookies at a special low price, the price increasing to its regular level after two weeks; the other store in the pair introduced the cookies at the regular price. Total sales of the cookies were computed for each store for six weeks from the time they were introduced. In 25 of the 36 pairs, the store which introduced the cookies at the regular price turned out to have sold more of them than the other store.

(a) Calculate a 95% confidence interval for the proportion of store pairs that report increased sales with the regular price.

(b) Consider the null hypothesis that discount pricing has no effect on sales, vs. the alternate hypothesis that discount pricing makes the targeted goods sell either more or less quickly. State these hypotheses more formally, and test whether the null or alternative hypothesis should be embraced, with Type I error 5%. Clearly state for what values of the test statistic the null hypothesis will be rejected, and whether you reject it in this case.

(c) Compute the associated p -value.